

REDIS 安装部署文档



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1 Redis 集群安装

序号	主机名	地址	安装目录	部署基础软件	备注
1	redis-01	192.168.4.24/24	/usr/local/redis-cluster	redis	
2	redis-01	192.168.4.25/24	/usr/local/redis-cluster	redis	
3	redis-03	192.168.4.26/24	/usr/local/redis-cluster	redis	
4	Node-10	192.168.4.20/24	/usr/local/redis-cluster	redis	
5	Node-11	192.168.4.21/24	/usr/local/redis-cluster	redis	
6	Node-12	192.168.4.22/24	/usr/local/redis-cluster	redis	

1.1操作环境

CentOS Linux release 7.7.1908 (Core)

1.2主机信息信息检查

1.2.1 网络相关检查

- 1) 检查确认主机名，与规划文档进行核对确认主机名是否需要修改。

检查：

```
[root@localhost ~]# hostnamectl
Static hostname: localhost.localdomain
Icon name: computer-vm
Chassis: vm
Machine ID: 8633a1bd91e447908a40c0df8eaec169
Boot ID: f293dbd6c7d642439394156c7cffe519
Virtualization: vmware
Operating System: CentOS Linux 7 (Core)
CPE OS Name: cpe:/o:centos:centos:7
```

```
Kernel: Linux 3.10.0-514.el7.x86_64  
Architecture: x86-64
```

修改:

```
[root@localhost /]# hostnamectl --static set-hostname redis-01
```

```
[root@sitpp1 ~]# hostnamectl
```

```
Static hostname: nginx-01  
Icon name: computer-vm  
Chassis: vm  
Machine ID: 0fb41200949e3c54b8d1dea4e83c3074  
Boot ID: b9d1ecb1799b4f34a32df70e168cbda0  
Virtualization: kvm  
Operating System: CentOS Linux 7 (Core)  
CPE OS Name: cpe:/o:centos:centos:7  
Kernel: Linux 3.10.0-1062.el7.x86_64  
Architecture: x86-64
```

2) 执行以下命令确认是否删除虚拟网络。

检查方法如下:

```
[root@localhost /]# ip addr list
```

```
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host  
        valid_lft forever preferred_lft forever  
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000  
    link/ether 52:54:00:12:34:56 brd ff:ff:ff:ff:ff:ff  
    inet 192.168.2.44/24 brd 192.168.2.255 scope global noprefixroute dynamic eth0  
        valid_lft 76872sec preferred_lft 76872sec  
    inet6 fe80::5054:00ff:fe00:0000/64 scope link  
        valid_lft forever preferred_lft forever
```

3) 时钟同步

查看时区

```
# timedatectl
```

```
Local time: Thu 2020-05-21 09:40:07 CST  
Universal time: Thu 2020-05-21 01:40:07 UTC  
RTC time: Thu 2020-05-21 01:40:06  
Time zone: Asia/Shanghai (CST, +0800)  
NTP enabled: n/a  
NTP synchronized: no  
RTC in local TZ: no  
DST active: n/a
```

设置时区

```
# timedatectl set-timezone Asia/Shanghai
```

配置时间同步

chrony 方式配置

```
vim /etc/chrony.conf
#添加以下内容
server 192.168.147.80 iburst
server ntp1.aliyun.com iburst
[root@localhost]# systemctl start chronyd.service
[root@localhost]# systemctl restart chronyd.service
[root@localhost]# systemctl status chronyd.service
[root@localhost]# systemctl enable chronyd.service
[root@localhost]# chronyc -n sources -v
```

4) SELINUX

```
[root@localhost /]# sed -i 's/SELINUX=enforcing/SELINUX=disabled/' /etc/selinux/config
[root@localhost /]# grep SELINUX=disabled /etc/selinux/config
SELINUX=disabled
[root@localhost /]# setenforce 0
```

5) 防火墙设置

firewall 关闭

```
[root@localhost /]# systemctl stop firewalld.service
[root@localhost /]# systemctl disable firewalld.service
```

1.2.2 磁盘相关检查

确认磁盘大小与规划大小是否一致，确认分区是否一致

```
[root@localhost /]# fdisk -l
```

1.2.3 CPU、内存检查

确认 CPU、内存与规划大小是否一致，确认分区是否一致

查看 CPU 个数

```
# cat /proc/cpuinfo | grep "physical id" | uniq | wc -l
[root@CICD-01 ~]# cat /proc/cpuinfo | grep "physical id" | uniq | wc -l
4
```

查看 CPU 核数

```
# cat /proc/cpuinfo | grep "cpu cores" | uniq
[root@CICD-01 ~]# cat /proc/cpuinfo | grep "cpu cores" | uniq
cpu cores      : 1
```

查看 CPU 型号

```
# cat /proc/cpuinfo | grep 'model name'| uniq
[root@CICD-01 ~]# cat /proc/cpuinfo | grep 'model name'| uniq
model name      : QEMU Virtual CPU
```

查看内存总数

```
#cat /proc/meminfo | grep MemTotal
[root@CICD-01 ~]# cat /proc/meminfo | grep MemTotal
MemTotal:      8008712 kB    //内存 8G
```

1.2.4 目录相关检查

检查介质目录

```
[root@localhost /]# ll /software
```

创建介质目录将相关介质放置该目录备用

```
[root@localhost /]# mkdir /software
```

1.3 安装 Redis

1.3.1 安装依赖包

1) 安装 GCC、RUBY、libyaml-devel

```
yum -y install gcc ruby
rpm -ivh libyaml-devel-0.1.4-11.el7_0.x86_64.rpm

[root@sgcs ~]# rpm -ivh libyaml-devel-0.1.4-11.el7_0.x86_64.rpm
warning: libyaml-devel-0.1.4-11.el7_0.x86_64.rpm: Header V3 RSA/SHA256 Signature, key ID f4a80eb5: NO KEY
Preparing...                               ##### [100%]
Updating / installing...
 1:libyaml-devel-0.1.4-11.el7_0             ##### [100%]
[root@sgcs ~]# rvm install 2.4.1
Searching for binary rubies, this might take some time.
Found remote file https://rvm_io.global.ssl.fastly.net/binaries/centos/7/x86_64/ruby-2.4.1.tar.bz2
Checking requirements for centos.
Installing requirements for centos.
Installing required packages: patch, autoconf, automake, bison, gcc-c++, libffi-devel, libtool, patch, readline-devel, sqlite-devel, zlib-devel, openssl-devel.....
Requirements installation successful.
ruby-2.4.1 - #configure
ruby-2.4.1 - #download
% Total    % Received % Xferd Average Speed   Time    Time     Time  Current
   Dload  Upload   Total   Spent    Left   Speed
 14 14.1M   14 2048k    0     0  28869      0  0:08:32  0:01:12  0:07:20 28129
```

安装 OPENSLL

```
rpm -Uvh openssl-1.0.2k-8.el7.x86_64.rpm openssl-devel-1.0.2k-8.el7.x86_64.rpm
openssl-libs-1.0.2k-8.el7.x86_64.rpm --nodeps --force
yum install openssl*
```

1.3.2 升级 Redis 集群 Ruby 版本

```
\curl -sSL https://raw.githubusercontent.com/wayneeseguin/rvm/master/binscripts/rvm-installer |
bash
```

```
[root@node-01 software]# \curl -sSL https://raw.githubusercontent.com/wayneeseguin/rvm/master/binscripts/rvm-installer |
bash
Downloading https://github.com/rvm/rvm/archive/master.tar.gz
Creating group 'rvm'
Installing RVM to /usr/local/rvm/
Installation of RVM in /usr/local/rvm/ is almost complete:

* First you need to add all users that will be using rvm to 'rvm' group,
  and logout - login again, anyone using rvm will be operating with `umask u=rwx,g=rwx,o=rx`.

* To start using RVM you need to run `source /etc/profile.d/rvm.sh`
  in all your open shell windows, in rare cases you need to reopen all shell windows.
* Please do NOT forget to add your users to the rvm group.
  The installer no longer auto-adds root or users to the rvm group. Admins must do this.
  Also, please note that group memberships are ONLY evaluated at login time.
  This means that users must log out then back in before group membership takes effect!
Thanks for installing RVM 🍷
Please consider donating to our open collective to help us maintain RVM.
💖 Donate: https://opencollective.com/rvm/donate
```

激活 W

```
[root@sgcs src]# source ~/.bashrc
[root@sgcs src]# source ~/.bash_profile
```

使用 source 让当前 shell 读入路径为 " /usr/local/rvm/scripts/rvm " (默认路径) 的 shell 文件并依次执行文件中的所有语句, 并重新执行刚修改的初始化文件, 使之立即生效, 而不必注销并重新登录, 输入命令 "source /usr/local/rvm/scripts/rvm ", 查看 rvm 中管理的所有 ruby 版本, 输入命令 "rvm list known" 进行查询, 如下图:

```
[root@redis-01 redis]# rvm list known
# MRI Rubies
[ruby-]1.8.6[-p420]
[ruby-]1.8.7[-head] # security released on head
[ruby-]1.9.1[-p431]
[ruby-]1.9.2[-p330]
[ruby-]1.9.3[-p551]
[ruby-]2.0.0[-p648]
[ruby-]2.1[.10]
[ruby-]2.2[.10]
[ruby-]2.3[.7]
[ruby-]2.4[.4]
[ruby-]2.5[.1]
[ruby-]2.6[.0-preview2]
ruby-head
```

选择一个版本进行安装, 所选择的版本不能低于 "2.2.2" 就可以, 输入命令 "rvm install 2.4.4" 进行安装, 如下图:

```

ruby-2.4.10 - #extracting rubygems-3.0.8.....
ruby-2.4.10 - #removing old rubygems.....
ruby-2.4.10 - #installing rubygems-3.0.8.....
ruby-2.4.10 - #gemset created /usr/local/rvm/gems/ruby-2.4.10@global
ruby-2.4.10 - #importing gemset /usr/local/rvm/gemsets/global.gems.....
.....
ruby-2.4.10 - #generating global wrappers.....
ruby-2.4.10 - #gemset created /usr/local/rvm/gems/ruby-2.4.10
ruby-2.4.10 - #importing gemsetfile /usr/local/rvm/gemsets/default.gems evaluated to empty gem list
ruby-2.4.10 - #generating default wrappers.....
ruby-2.4.10 - #adjusting #shebangs for (gem irb erb ri rdoc testrb rake).
Install of ruby-2.4.10 - #complete
Ruby was built without documentation, to build it run: rvm docs generate-ri

```

rvm docs generate-ri

使用刚才安装完成的 Ruby 版本，输入命令 " rvm use 2.4.4 "

1.3.3 安装

```

tar -xvf redis-4.0.8.tar
cd redis-4.0.8
make MALLOC=libc
make install PREFIX=/usr/local/redis-cluster

```

创建 Redis 实例

```

cd /usr/local/redis-cluster
mkdir re7000

进入实例编写配置文件
vim redis.conf

port 7000
bind 192.168.4.25
cluster-enabled yes
cluster-config-file nodes.conf
cluster-node-timeout 5000
appendonly yes
daemonize yes
masterauth "Sgaizx!"
requirepass "Sgaizx!"
logfile "/usr/local/redis-cluster/redis7000.log"
# Generated by CONFIG REWRITE
dir "/usr/local/redis-cluster/re7000"

```

注意：端口号要与文件夹端口号保持一致。剩余文件夹均要创建配置文件修改对应端口号即可。

加入集群时，可能报错，把 Redis 配置文件中密码配置删掉重启 Redis 在试下。

将 redis 编译后源码文件 src 文件夹下 redis-server、redis-trib.rb、redis-cli 文件拷贝至
/usr/local/redis-cluster

cp redis-cli redis-server redis-trib.rb /usr/local/redis-cluster/

编写 Redis 实例启动脚本(6 个节点都编写)

```
vim start.sh
#!/bin/bash
cd /usr/local/redis-cluster/re7000/ && ../redis-server ./redis.conf
启动 redis
./start.sh
```

搭建 Redis 集群

安装 Redis 接口

```
gem install redis
```

报错:

```
[root@redis-01 redis]# gem install redis
ERROR: While executing gem ... (Gem::Exception)
Unable to require openssl, install OpenSSL and rebuild Ruby (preferred) or use non-HTTPS sources
```

解决方法: <https://blog.csdn.net/feinifi/article/details/78251486>

安装 Redis 集群

```
./redis-trib.rb create --replicas 1 192.168.4.24:7000 192.168.4.25:7000
192.168.4.26:7000 192.168.4.20:7000 192.168.4.21:7000 192.168.4.22:7000
```

```
[root@redis-01 redis-cluster]# ./redis-trib.rb create --replicas 1 192.168.4.24:7000
192.168.4.25:7000 192.168.4.26:7000 192.168.4.20:7000 192.168.4.21:7000
192.168.4.22:7000
>>> Creating cluster
>>> Performing hash slots allocation on 6 nodes...
Using 3 masters:
192.168.4.24:7000
192.168.4.25:7000
192.168.4.26:7000
Adding replica 192.168.4.21:7000 to 192.168.4.24:7000
Adding replica 192.168.4.22:7000 to 192.168.4.25:7000
Adding replica 192.168.4.20:7000 to 192.168.4.26:7000
M: adaaa7e839f5b719acfbe44fa0ea6222d255b9de 192.168.4.24:7000
slots:0-5460 (5461 slots) master
M: a51a6ef6e557bd29ed0a8a46290768770b0c1aca 192.168.4.25:7000
slots:5461-10922 (5462 slots) master
M: bd689f3d273ddfc2668dc0733c5626bd3afc3e17 192.168.4.26:7000
slots:10923-16383 (5461 slots) master
S: fd77d2b3a0574762d43d5af41f1ea453df2da168 192.168.4.20:7000
replicates bd689f3d273ddfc2668dc0733c5626bd3afc3e17
S: 3e44f1283bf8affecfc7b1c25e19aaa7ee0d504a 192.168.4.21:7000
replicates adaaa7e839f5b719acfbe44fa0ea6222d255b9de
S: 59fe8d1617158ab328cc748db98811a9df1438fb 192.168.4.22:7000
replicates a51a6ef6e557bd29ed0a8a46290768770b0c1aca
Can I set the above configuration? (type 'yes' to accept): yes
```

查看 Redis 集群状态

```
./redis-cli -p 7000 -h 192.168.4.25 -a Sgaizx! cluster info
```

```
cluster_state:ok
cluster_slots_assigned:16384
cluster_slots_ok:16384
cluster_slots_pfail:0
cluster_slots_fail:0
cluster_known_nodes:6
cluster_size:3
cluster_current_epoch:10
cluster_my_epoch:10
cluster_stats_messages_ping_sent:9085
cluster_stats_messages_pong_sent:7603
```

```
cluster_stats_messages_fail_sent:14
cluster_stats_messages_auth_req_sent:5
cluster_stats_messages_update_sent:2
cluster_stats_messages_sent:16709
cluster_stats_messages_ping_received:7598
cluster_stats_messages_pong_received:8060
cluster_stats_messages_fail_received:7
cluster_stats_messages_auth_ack_received:2
cluster_stats_messages_received:15667
```

1.3.4 Redis 开机自启

`vim /etc/systemd/system/redis.service`

#添加以下命令

```
[Unit]
Description=redis-server
After=network.target

[Service]
Type=forking
ExecStart=/usr/local/redis-cluster/redis-server
/usr/local/redis-cluster/re7000/redis.conf
PrivateTmp=true

[Install]
WantedBy=multi-user.target
```

授权

```
systemctl enable redis.service
```